(a) Privileged teacher supervision

Simulation ground truth

Object pose & velocity

Goal orientation & error

Proprioception

Hand state + prev. action

Teacher network

Actor MLP

Critic MLP

Teacher action

Simulator

Reward / PPO

Frozen teacher

Forward

Supervision

(b) Distill a goal-conditioned student

Language

Semantic agent

Public skill selection

SigLIP2 skill bank

Goal-bank vector

g

Sampled during training

Goal projection Linear + tanh

z_g

RGB observation

Observed RGB (simulation / camera)

Frozen pose CNN

Pose encoder MLP

Training input: noisy sim pose

Hand state

Proprioception + history

Concat

Student policy

Recurrent encoder

Actor MLP

Action a_S

Distillation

Action MSE

State MSE (training)

Training only

State decoder MLP

Deploy student

(c) Closed-loop execution

Live agent

Student policy

Initial view
Target not directly identifiable

Target bound
Requested view

RGB pose feedback
Real-hand results reported in Q4

Language instruction

Present the face marked T

New RGB